

Sections: All CS and SE

Probability and Statistics (MT-2005)

Homework #01 (c)

Data Visualization (Quantitative Data)

Question 1

Histogram Creation

A survey of 50 students' ages shows the following frequency distribution:

- 15-20 years: 12 students
- 21-25 years: 18 students
- 26-30 years: 10 students
- 31-35 years: 6 students
- 36-40 years: 4 students

Guidance: Draw a histogram to represent the distribution of ages. The x-axis will represent age ranges, and the y-axis will represent the number of students.

Python Task: Use the following Python code to create the histogram:

```
import matplotlib.pyplot as plt

age_groups = ['15-20', '21-25', '26-30', '31-35', '36-40']
students = [12, 18, 10, 6, 4]

plt.bar(age_groups, students)
plt.xlabel('Age Groups')
plt.ylabel('Number of Students')
plt.title('Age Distribution of Students')
plt.show()
```

Question 2

Frequency Polygon

Concept: A frequency polygon is a graphical representation of a frequency distribution. It is formed by plotting the midpoints of the classes on the x-axis and the frequency on the y-axis, then connecting the points with straight lines.

Numerical Problem:

Given the following frequency distribution of the number of hours studied by students, plot the frequency polygon:

Hours Studied	Frequency
0 – 5	8
5 – 10	12
10 – 15	15
15 – 20	10
20 – 25	5

Python Task:

1. Create a Python program to compute the midpoints of each class.
2. Plot the frequency polygon using `matplotlib` and mark the midpoints on the graph.

```
import matplotlib.pyplot as plt

# Data
classes = [(0, 5), (5, 10), (10, 15), (15, 20), (20, 25)]
frequencies = [8, 12, 15, 10, 5]

# Compute midpoints
midpoints = [(c[0] + c[1]) / 2 for c in classes]

# Plot frequency polygon
plt.plot(midpoints, frequencies, marker='o', linestyle='—',
         color='b', label='Frequency Polygon')
plt.title('Frequency Polygon')
plt.xlabel('Midpoint of Class Intervals')
plt.ylabel('Frequency')
plt.grid(True)
plt.legend()
plt.show()
```

Question 3

Frequency Curve

Concept: A frequency curve is a smooth curve drawn through the points of a frequency polygon. It represents the continuous data distribution.

Numerical Problem:

Using the frequency distribution from the previous question, plot a frequency curve. Assume that the data follows a normal distribution for simplicity.

Python Task:

1. Fit a normal distribution to the data and plot the frequency curve using `scipy` and `matplotlib`.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Data (from previous problem)
data = [0, 5, 10, 15, 20] # Simplified example
frequencies = [8, 12, 15, 10, 5]

# Fit a normal distribution to the data
mu, std = norm.fit(data)

# Plot histogram
plt.hist(data, bins=5, density=True, alpha=0.6, color='g')

# Plot the frequency curve (normal distribution)
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = norm.pdf(x, mu, std)
plt.plot(x, p, 'k', linewidth=2)

plt.title('Frequency Curve')
plt.xlabel('Data')
plt.ylabel('Probability Density')
plt.grid(True)
plt.show()
```

Question 4

Ogive

Concept: An ogive is a cumulative frequency curve. It helps visualize the cumulative distribution of the data.

Numerical Problem:

For the same frequency distribution:

Hours Studied	Frequency
0 – 5	8
5 – 10	12
10 – 15	15
15 – 20	10
20 – 25	5

Construct an ogive and explain its significance.

Python Task:

1. Calculate cumulative frequencies and plot the ogive.

```
import matplotlib.pyplot as plt
import numpy as np

# Data (same as previous problems)
classes = [(0, 5), (5, 10), (10, 15), (15, 20), (20, 25)]
frequencies = [8, 12, 15, 10, 5]

# Calculate cumulative frequencies
cumulative_frequencies = np.cumsum(frequencies)

# Compute the upper boundaries for each class
upper_boundaries = [c[1] for c in classes]

# Plot ogive
plt.plot(upper_boundaries, cumulative_frequencies, marker='o',
linestyle='-', color='r', label='Ogive')
plt.title('Ogive')
plt.xlabel('Upper Class Boundaries')
plt.ylabel('Cumulative Frequency')
plt.grid(True)
plt.legend()
plt.show()
```